

Total No. of Questions : 8]

[Total No. of Printed Pages : 2

Roll No

CS-3003-CBGS

B.E., III Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Digital Circuit and Design

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.
iii) Sketch the diagram neatly, if any.

1. a) Explain multiple output switching function. 7
b) State and prove basic law of Algebra. 7
2. a) What is Universal gate? Implement AND, OR, NOT gate using NAND gates. 7
b) Design and draw full adder circuits. 7
3. a) Explain working of PMOS, NMOS and CMOS logic. 7
b) Explain astable multivibrator. 7
4. a) Explain counters. Also differentiate between synchronous and asynchronous counter. 7
b) Write short note on shift registers. 7
5. a) What is the need of analog to digital converter? In detail explain any one digital converter. 7
b) Write short note on Sample and hold circuits. 7

CS-3003-CBGS

PTO

[2]

6. a) Simplify the Boolean function using K-map
 $F(A, B, C, D) = \sum(0, 1, 2, 4, 5, 6, 8, 9, 10, 12, 13, 14)$ 7
b) Explain BCD adder. 7
7. a) Explain Schmitt trigger circuit. 7
b) Explain semiconductor memories. 7
8. Write short notes on- 14
i) De-Morgan's law
ii) Latch (NAND and NOR)
iii) Astable multivibrator
iv) Ring counter
